



Global Quantum + AI Challenge 2026

Enterprise Challenge Statement

1. Challenge Title

Quantum Solvers: Enhancing Predictive Aerodynamic Modeling Capabilities

2. Executive summary

The ability to accurately predict aerodynamic flows is a key capability required within the aerospace sector. This ability requires the efficient solution of Partial Differential Equations (PDEs) utilising state of the art High Performance Computing to resolve complex multi-scale problems of many millions of degrees of freedom. Airbus is seeking more efficient ways to solve PDEs so novel product solutions can be identified which meet or exceed the challenging quality and environmental requirements of their customers.

Being able to accurately predict the aerodynamic forces at the edge of the aircraft operating envelope is required for sizing of the primary structure. It can also give critical information on fuel consumption at cruise conditions. Due to the limited capabilities of modelling methods and computational resources; there is still a reliance on wind tunnel testing at these extreme conditions of high acceleration and Mach number. As physical testing is expensive and impacts the lead time for prototyping, the question is: Can we find alternative modelling solutions which increase modelling and hence design capabilities?

Quantum computing offers the potential to resolve flow problems with a significantly greater number of degrees of freedom than classical computing techniques. This, in turn, provides the means to increase accuracy (through, for example, greater flow resolution) of aerodynamic predictions, overcoming some of the limitations common to conventional PDE solvers.

The Challenge: We challenge participants to demonstrate a potential advantage over classical techniques by simulating a canonical fluid dynamics test case: the Taylor-Green vortex in a streamwise flow. Although this is a mathematical idealisation, the physics that it represents is found in many real world flows from geophysical fluid dynamics to vortex interactions in aerospace. Because this problem features a known exact analytical solution for how the vortex decays over time, it provides a perfect benchmark to rigorously assess both computational fidelity and time-to-solution.

3. Industrial and Operational Context

The opportunity offered by Quantum Computing is to realise an alternative more efficient simulation paradigm which increases our ability to explore design options in a timely way and only requires physical testing in the final validation phase.

3.1 Classic methods for solver

Fluid motion is governed by the Navier–Stokes equations [1], a set of coupled and nonlinear partial differential equations derived from the basic laws of conservation of mass, momentum and energy.

The equations are currently typically solved using discretisation techniques such as Finite Volume [2], Discontinuous Finite Elements [3], or alternatively the Lattice Boltzmann method [4]. Using a discrete numerical approach enables the initial continuous problem to be mapped to a linear system of equations with a large sparse matrix to be inverted. The matrix inversion step makes those methods extremely heavy for application to realistic models at a high accuracy level. To solve nonlinear equations a time stepping scheme can be used together with a linear solver.

3.2 Quantum algorithm approaches for solvers

Quantum mechanical processes are inherently unitary and linear. However, the fluid mechanical processes of interest are nonunitary (dissipative) and nonlinear. Also our challenge case, the convecting Taylor Green Vortex in 2D being one of the simplest examples of a nonlinear, nonunitary flow. The key algorithmic challenge therefore is to map the nonlinear fluid problems onto unitary quantum hardware. Progress has been made on quantum algorithms for (a) solving linear equations, (b) linearising nonlinear systems of equations to make them amenable to a linear solver and (c) managing the nonunitary aspect. Examples are given below.

- a) One of the earliest and well-known quantum algorithms for solving linear systems of equations is that of Harrow, Hassidim, and Lloyd [5]. Subsequently an improved complexity with respect to precision was obtained in the quantum algorithm of Childs et al. [6].
- b) A number of approaches have been adopted in the literature for ‘linearising’ nonlinear PDEs making them amenable to quantum solution through linear solvers. These include, amongst others, a mean-field approach [7], Carleman linearisation [8, 9, 10] and a Liouville embedding [11]. With the adoption of a

time stepping procedure such as Euler forward differencing, the problem is reduced to a linear system of algebraic equations.

- c) Strategies to overcome the nonunitary issue include, amongst others, *Unitary Dilation* (i.e., block encoding) where a nonunitary linear system of interest is embedded into a larger, unitary matrix [12] and the *Linear Combination of Unitaries* where a nonunitary linear system of interest is written as a sum of unitary matrices [13].
- d) More recently, quantum machine learning (QML) frameworks have also been proposed to learn such nonlinearities directly. By training variational quantum circuits (VQC) to approximate the collision operator in the lattice Boltzmann method, these models can replicate complex nonlinear dynamics through unitary evolution [14, 15].

The methods above result in very large matrices to resolve the nonunitary and nonlinear issues and convergence at higher Reynolds numbers is not guaranteed. However, since FTQC hardware does not yet exist, algorithms must be verified using existing hardware which limits the range of problems that can be considered.

4. Challenge Objective

The goal of this challenge is to solve the 2D Convecting Taylor-Green Vortex. We invite participants to explore the possible spectrum of methodologies, ranging from native quantum algorithms to quantum-inspired classical approaches. While we acknowledge the current limitations of quantum hardware, recent advancements in Fault-Tolerant Quantum Computing (FTQC) offer a potential path towards industrial usefulness. As a consequence, we also encourage the investigation of algorithms designed for future fault-tolerant architectures, even if the necessary hardware is not yet operationally available. Simultaneously, Airbus maintains a strategic interest in all quantum solutions including quantum-inspired methods that can in principle provide advantages today.

Proposals will be evaluated based on their technical rigor, quality, and innovative spirit. Each approach, from quantum-inspired to different quantum algorithms, will be assessed on its own merits relative to its specific computational paradigm.

Quantum Algorithm: Develop a solver utilizing a quantum algorithm.

While heuristic approaches such as Variational Quantum Algorithms (VQAs), Physics-Informed Neural Networks (PINNs), and general Deep Learning techniques are prominent in the Noisy Intermediate-Scale Quantum (NISQ) era, they often face scalability and noise-floor limitations. Therefore, we also encourage methods that offer provable algorithmic scaling and are designed for the high-precision requirements of fault-tolerant architectures.

Participants are tasked with developing such a solver, which can be run on today's quantum hardware or future fault-tolerant machines. Suitable methodologies are not limited but may include one of the following approaches:

- Finite Volume or Lattice Boltzmann Methods (LBM)
- Schrödingerization
- Quantum Linear System Algorithms (QLSA)

Participants are also invited to choose quantum or classical AI methods that augment those approaches in order to enhance the performance of the proposed quantum solvers.

Tensor Network: Develop a solver based on Tensor Networks (TN) implemented within a Finite Volume (FV) method framework.

This track explores TN, a technique that acts as "intelligent compression" for complex physical data. For this challenge, we are interested in applying TN specifically within a FV framework. Unlike standard methods that discretize points on a grid, here you should represent the cell averages of the fields. At the same time the algorithm must be able to compute the relative field at the volume interface while the temporal evolution is driven by calculating the fluxes across volume faces rather than simple grid-point updates.

4.1 Scaling with respect to Reynolds Number

The Taylor-Green Vortex is a canonical benchmark to evaluate the accuracy and stability of numerical solvers. In particular, in the convecting Taylor-Green vortex, both diffusion (coming from the viscous diffusion) and the dispersion and dissipation terms (coming from the nonlinear advection) contribute to the numerical accuracy of the solver (see Eqs in the following).

The primary objective is to quantify how the algorithm scales, in terms of both runtime and accuracy, as the Reynolds number increases. Grid resolution must increase with Reynolds number to correctly model the physics. Specifically, we aim to have a plot

characterizing the obtained (or estimated) time-to-solution and the numerical error across a range of Reynolds. The analysis should determine:

- **Time-to-solution:** how the runtime cost grows with Re
- **Memory Requirement:** how the memory (or number of qubits) complexity scales with Re
- **Error Scaling:** how the stability of the method holds up with increasing Re. In particular, here we are interested in the (i) L2 error norm: the global error between the simulated velocity field and the exact analytical solution, or the (ii) kinetic energy decay.

The participants may wish to compare their results with a state of the art of the classical solver.

5. Problem Definition

The convecting Taylor-Green Vortex problem is governed by the **two-dimensional incompressible Navier-Stokes** equations for a Newtonian fluid with constant density (ρ) and kinematic viscosity (ν); where the conservation of mass (continuity) is

$$\nabla \cdot \mathbf{u} = 0 \implies \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

and momentum is given by

$$\begin{aligned} \frac{\partial u}{\partial t} + \underbrace{u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}}_{\text{Non-linear Advection}} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \underbrace{\nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)}_{\text{Viscous Diffusion}} \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \end{aligned}$$

$\mathbf{u}(x, y, t) = [u(x, y, t), v(x, y, t)]$ is the velocity field, p is the pressure and t is time.

5.2 Initial Conditions

The flow is initialized at time $t = 0$ with a smooth, deterministic two-dimensional vortex structure. The velocity components are defined as:

$$u(x, y, 0) = U_c + V_0 \sin\left(\frac{x}{L}\right) \cos\left(\frac{y}{L}\right)$$

$$v(x, y, 0) = V_c - V_0 \cos\left(\frac{x}{L}\right) \sin\left(\frac{y}{L}\right)$$

Where V_0 is the vortex intensity (i.e. how fast the fluid is spinning) while U_c and V_c are the convection velocity components (i.e. the x , y components of the streamflow that transport the vortex in the simulation domain). L is the characteristic length of the vortex. By applying the divergence operator to the momentum equation we obtain the Poisson equation:

$$\nabla^2 p = -\nabla \cdot [(\mathbf{u} \cdot \nabla) \mathbf{u}]$$

from which we derive the corresponding pressure field at the initial conditions:

$$p(x, y, 0) = p_0 + \frac{\rho V_0^2}{4} \left[\cos\left(\frac{2x}{L}\right) + \cos\left(\frac{2y}{L}\right) \right]$$

5.3 Exact Analytical Solution

A key feature of the 2D configuration is the existence of an exact closed-form solution for all $t > 0$. The temporal evolution is characterized by a simple exponential decay governed by ν :

$$u(x, y, t) = U_c + V_0 \sin\left(\frac{x - U_c t}{L}\right) \cos\left(\frac{y - V_c t}{L}\right) e^{-2\nu t/L^2}$$

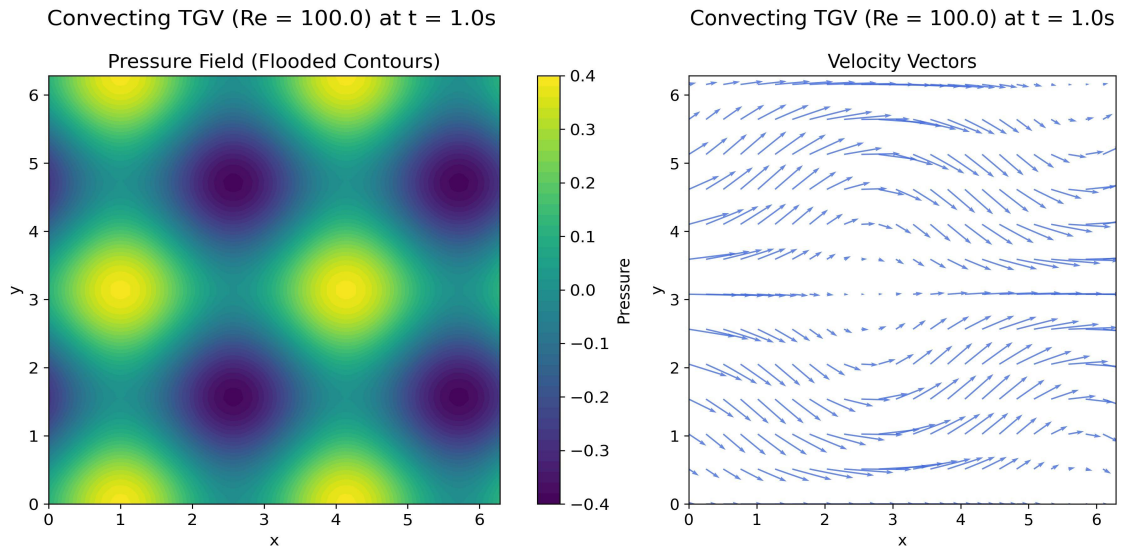
$$v(x, y, t) = V_c - V_0 \cos\left(\frac{x - U_c t}{L}\right) \sin\left(\frac{y - V_c t}{L}\right) e^{-2\nu t/L^2}$$

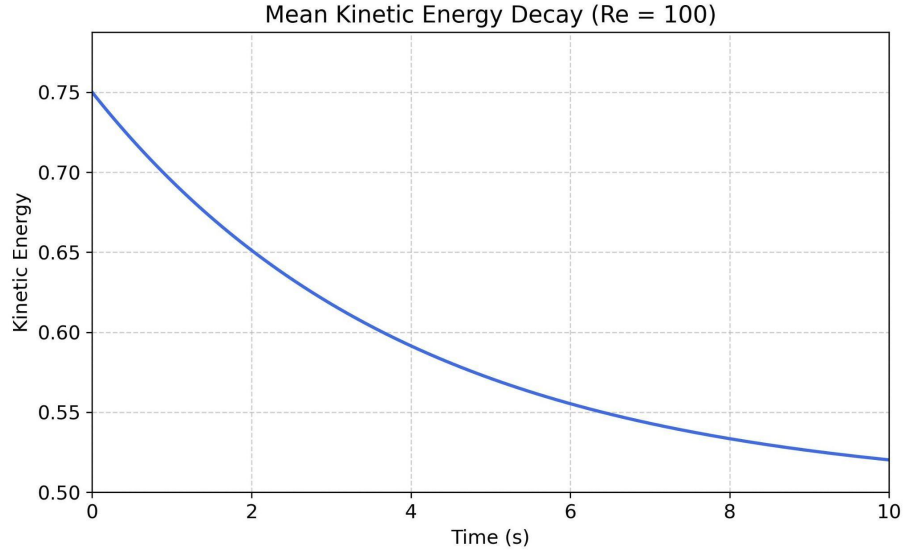
This allows for the precise calculation of error without relying on other reference data.

6. Data and Resources

Parameter	Value
Domain Length (L)	2π
Vortex Velocity (V_0)	1.0
Convection Velocity in X (U_c)	1.0
Convection Velocity in Y (V_c)	0.0
Density (ρ)	1.0
Reynolds number (Re)	10, 100, to the maximum required to demonstrate possible advantage.
Viscosity (ν)	$= V_0 L / Re$
Background Reference pressure (p_0)	0.0

Illustrations of the analytic solution velocity and pressure fields, together with the mean kinetic energy decay, at time $t = 1$ s for the above input parameter settings with $Re = 100$ is given below.





7. References

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